



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Date: August 12th 2026



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Summary of methodologies:

- **Data collection and Cleaning** - Collected the data from SpaceX launch data API. Then formatted the data and cleaned it by removing duplicates and missing values as well as creating target variables. Also executed various SQL queries
- **Exploratory Data Analysis** - Performed data analysis on the launch and payload data, while comparing various variable and creating graphs which showed success rate of these launch sites, orbit etc.
- **Data Visualization** - Created charts with the help of Matplotlib, Seaborn etc. Used folium to show the markers, polyline or distance, and analyzed proximity to coastlines, cities , railways etc.
- **Dashboards and Interactive Visualization** - Created interactive dashboard by using dash and showcased the success of the different launch sites and and payload masses.
- **Model Building and Evaluation** - Created ML models and implemented K-NN, SVM, Decision Trees. Also used GridSearchCV to find the best hyperparameters and checked and compared each models accuracy score.

Executive Summary

Summary of Result:

- Data Insights - By the use of different methods various factors were identified that influence the landing of Falcon 9.
- Findings - Launch site and Payload mass impact the landing success. Also, Falcon 9 landing success rate increases over time.
- Model Performance - The performance of both Support Vector Machines and K-Nearest Neighbors is 83.33% accuracy, and the decision tree model has the highest accuracy with a score of 94.44%. Thus the decision tree model is the best model amongst all.

Introduction

Project background and context -

In this final project, we aim to predict the successful landing of the Falcon 9 first stage. SpaceX is one of the few firm who have tried and succeeded in developing a reusable rocket. This approach allows SpaceX to advertise rocket launches with a significantly lower costs. By predicting the success of the first stage we can understand the factors that affect the rockets reusability and estimate launch costs.

Problems you want to find answers -

- Which factors influence Falcon 9 first stage landing success and how does the success evolve with respect to time?
- How can we accurately predict with machine learning and which ML model shows the best result.
- How does factor relate to each other like orbit, payload and launch site.

Section 1

Methodology

Methodology

Executive Summary: This projects implements a technical approach from collecting data via the SpaceX API, conducting exploratory data analysis, creating interactive visualizations and finally conducting predictive modeling.

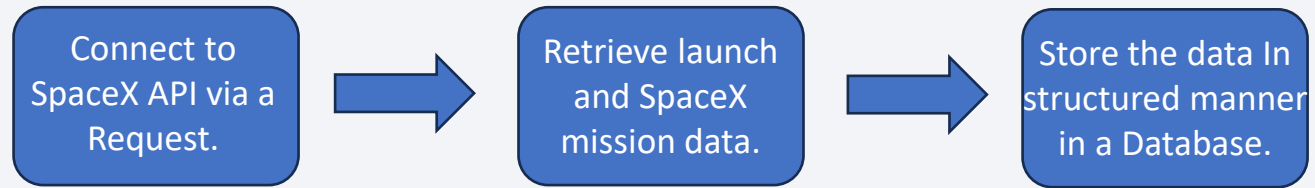
- **Data collection methodology:** Data was collected from the SpaceX API and was converted from the JSON format. This data provided records or Payload mass, orbit and launch sites.
- **Perform data wrangling:** The raw data was cleaned, formatted and processed with the help of Pandas and Numpy including handling missing values and creating features for better analysis.
- **Perform exploratory data analysis (EDA) using visualization and SQL:** Visualized graphs which showcased launch success rate, relation between launch success and payload mass or orbit. Carried out SQL queries to retrieve data and derive insights.

Methodology

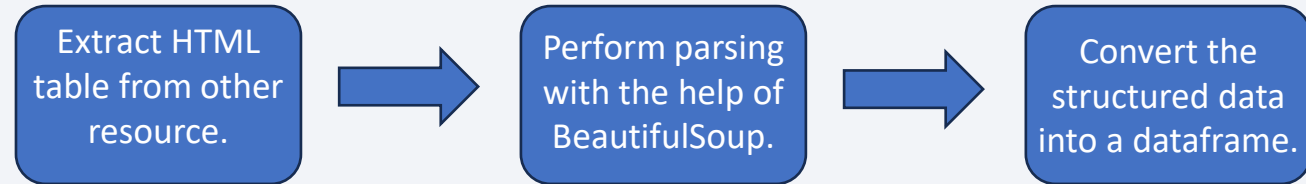
- **Perform interactive visual analytics using Folium, Plotly, and Dash:** Created interactive maps through folium to display launch sites, distance lines and proximity to various points. Moreover, developed interactive dashboard that showcased the success rate of launch site and their relation, with the size of payload mass
- **Perform predictive analysis using classification models:** Built and evaluated various models such as Support Vector Machines, K-Nearest Neighbors and Decision tree. Employed GridSearchCV to find the best hyperparameters. Lastly, measured the accuracy of each model, and found the most optimal one.

Data Collection

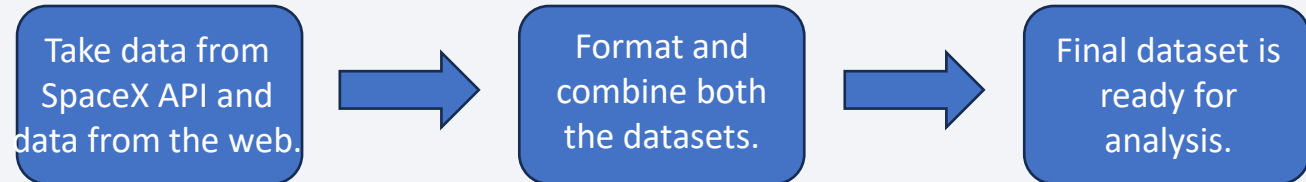
Step 1: SpaceX API -



Step 2: Web Scraping -

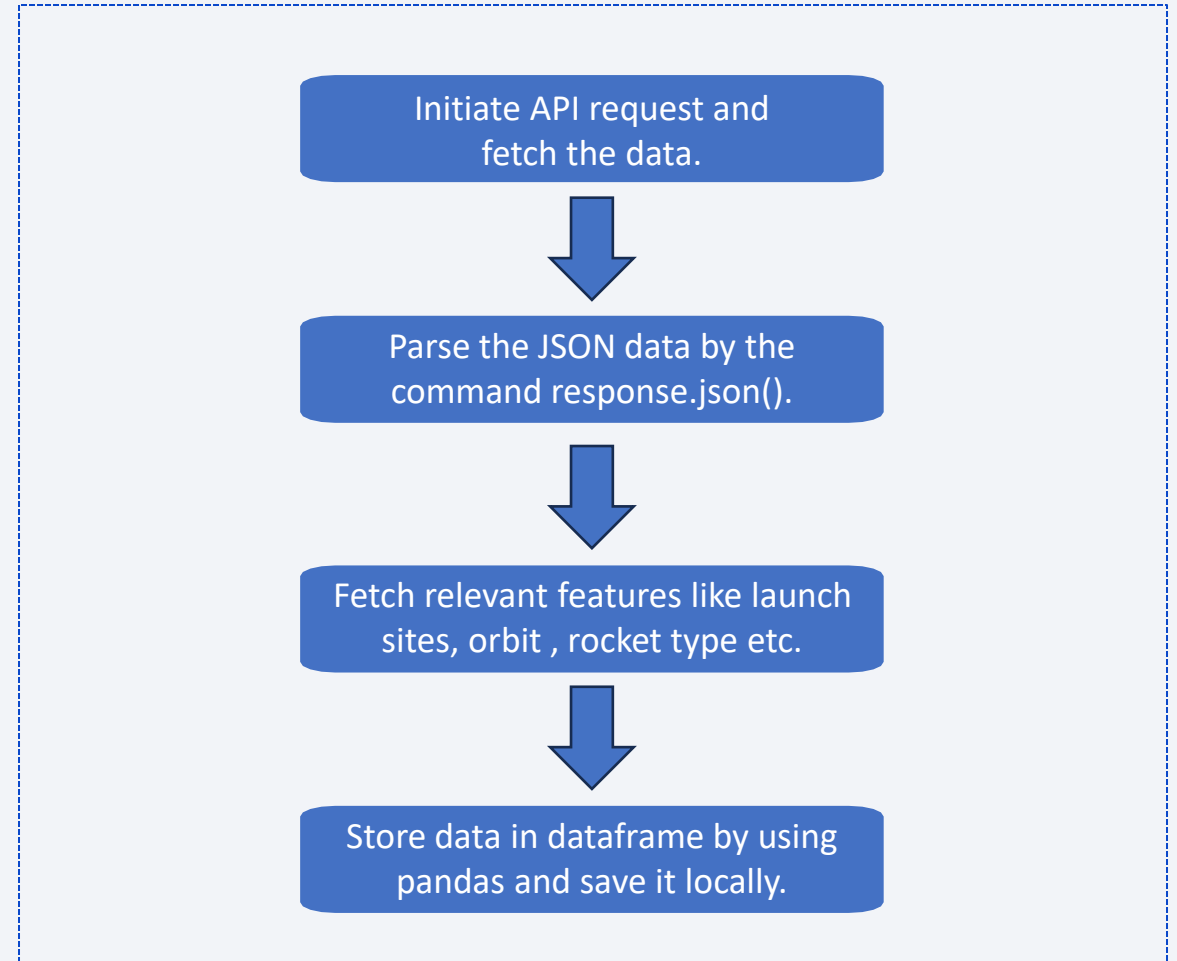


Step 3: Data Wrangling -



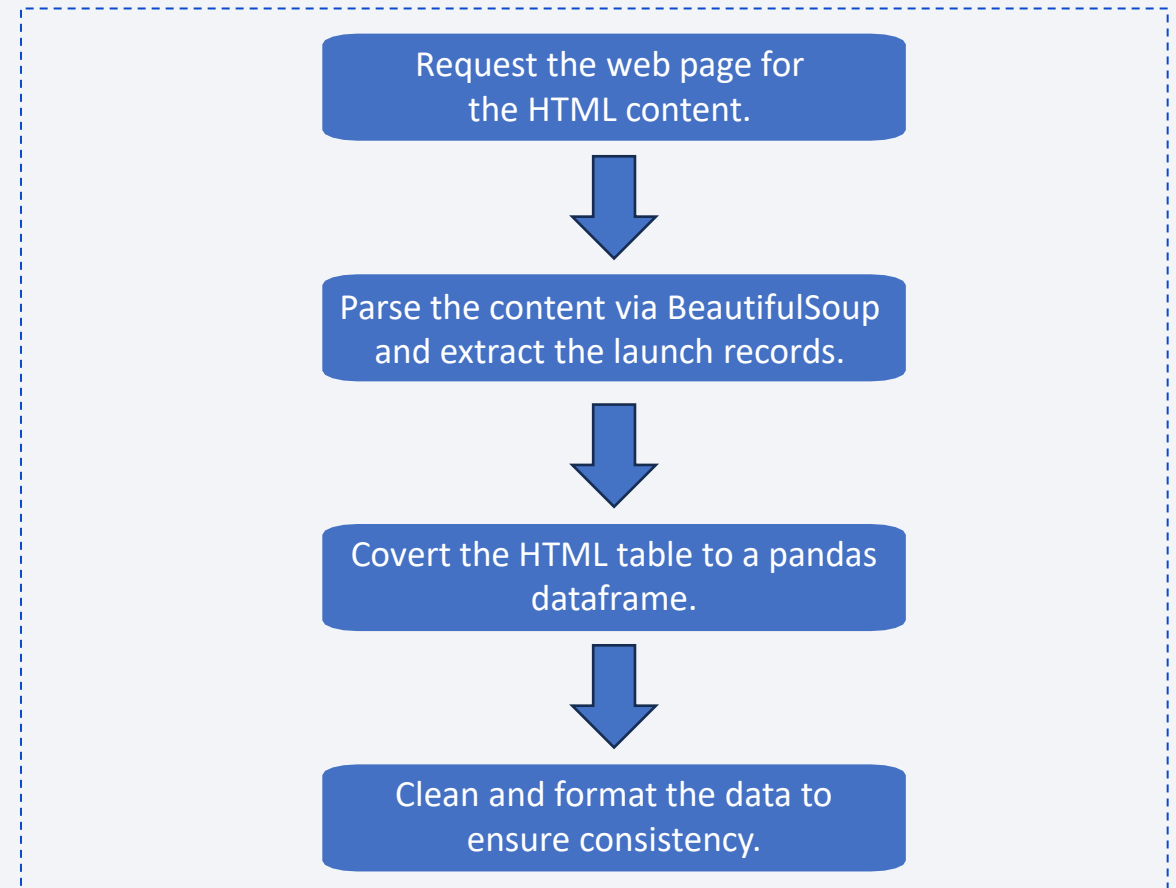
Data Collection – SpaceX API

1. **SpaceX API request:** Initiate a request to SpaceX API to get the data by importing the request library. Link to the `APU` – <https://api.spacexdata.com/v4/launches>
2. **Parsing:** Next the data from the API in the form of json is parsed. The necessary feature such as booster versions, rocket, launch site are extracted.
3. **Storing Data:** The is converted form json to format which pandas can read and is converted to a dataframe.



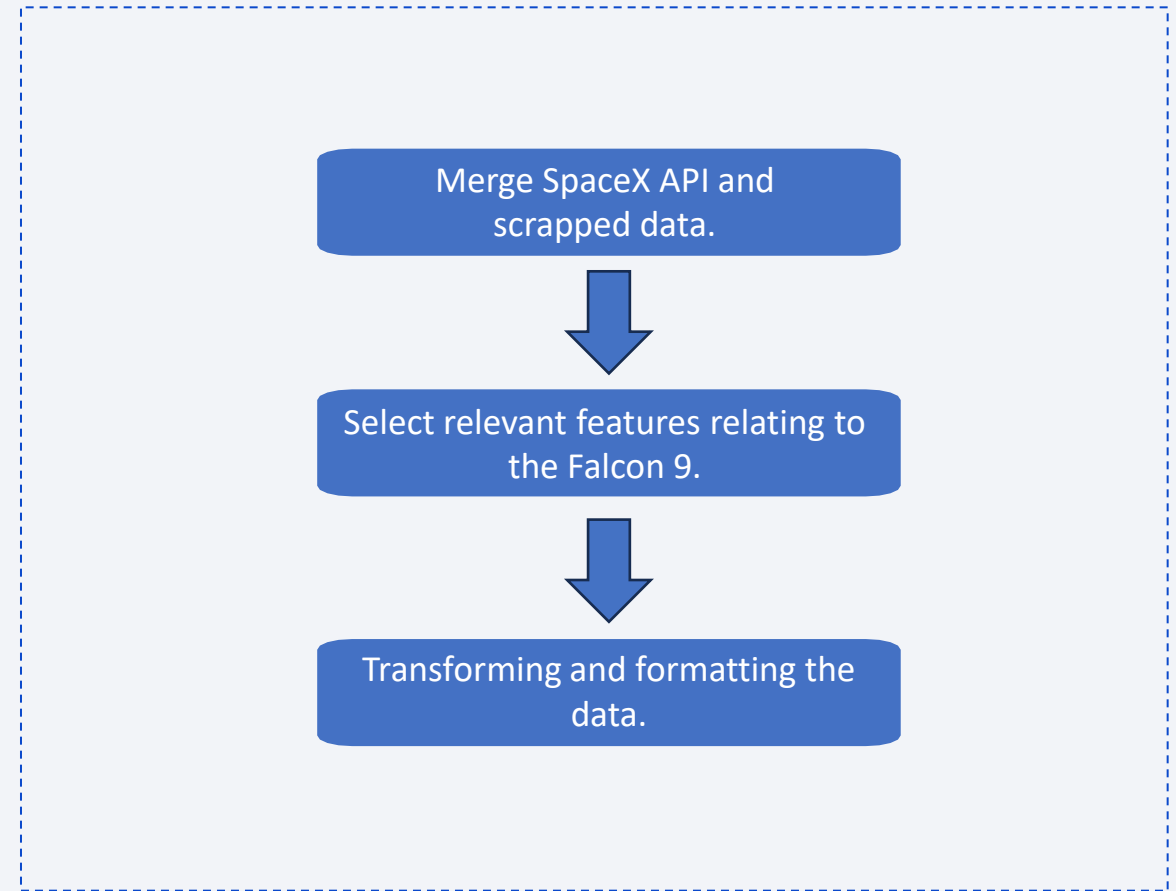
Data Collection - Scraping

1. **Initiate Web Scraping** - First use the python request library to fetch the HTML data. URL - [`https://en.wikipedia.org/wiki/List\\_of\\_Falcon\\_9\\_and\\_Falcon\\_Heavy\\_launches`](https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches)
2. **Parsing HTML** - Next use the BeautifulSoup to parse the HTML table and extract data regarding Falcon 9 launches.
3. **Storing the data** - Convert and format the extracted data into a dataframe through the Pandas Library



Data Wrangling

1. **Combine Datasets** - The data sets form the SpaceX API and web-scraped data.
2. **Selecting Relevant Features** - Only those variable or features are selected relating to the Falcon 9 such as rocket and payloads.
3. **Transform and Cleaning** - Handle missing value, convert data types, standardize text, and normalizing numerical values.



Exploratory Data Analysis with Data Visualization

Exploratory Data Analysis (EDA) is process where data is visualized and examined to identify the patterns, relationships, trends between different variables. The different types of visualization –

1. **Scatter Plot** - Scatter plot are graph which allow us to explore the relationship between 2 numerical variables. It identifies correlation and how one variable changes in relation to the other. The scatter plot was used to show the relation between payload mass and launch site and to prove whether it influence its success rate.
2. **Bar chart** - Bar charts compare the value across various categorical variables such as comparing the success rate of each launch site. Bar charts provide a clear comparison of frequencies between different categories.
3. **Pie chart** - Pie charts display the proportion of different variable, for e.g. in This project it was used to landing success rate across different launch sites.
4. **Histograms** - Histogram illustrate the distribution of numeric data or numeric variables. By using a, histogram we can see the spread and tendency of data such as in the case of payload masses.

EDA with Data Visualization

5. **Line charts** - Line charts are good way to show trends over time and analyze patterns during a specified period. Line charts showed how the success rate changed over the year, thus showing how they improved.
6. **Heatmaps** - These graphical representation show the correlation matrix or correlation coefficient of different numerical variables. These heatmaps help us to know the correlation between different features and selecting those feature with the highest correlation.
7. **Box Plot** - These are used to display the distribution of the data across their quartiles. Box plots shows the spread of data compares the distribution as well as at the same time highlights the outliers.
8. **Folium map** - With the help of folium we can conduct spatial and geographical analysis. By plotting the latitude and longitude of SpaceX launch sites along with other markers we can check its proximities and distances which could tell us important details.

URL: [https://github.com/hamzaqdas7-design/IBM-final-project/blob/main/edadataviz%20\(1\).ipynb](https://github.com/hamzaqdas7-design/IBM-final-project/blob/main/edadataviz%20(1).ipynb)

EDA with SQL

Selecting and Filtering Queries:

- Calculated total number of launches.
- Filtering launch sites which begin with CCA
- Retrieving drone-ship landings by payload which lie between 4000 and 6000.
- Filter drone ship failures in 2015

Aggregating and Statistical Queries :

- Total of NASA CRS Payload.
- Average of F9 v1.1 payload

Minimum and Maximum Queries

- First successful ground landing
- Maximum Payload Booster

Grouping and Counting Queries:

- Counting mission outcomes like amount of success and failures.
- Ranking landing outcome or counting each landing outcome within a specified period of time.

Subqueries:

- There is a subquery to find the booster associated with the maximum payload mass
- Subqueries used to perform analysis and further selection in large datasets.

URL: [https://github.com/hamzaqdas7-design/IBM-final-project/blob/main/jupyter-labs-eda-sql-coursera_sqlite%20\(1\).ipynb](https://github.com/hamzaqdas7-design/IBM-final-project/blob/main/jupyter-labs-eda-sql-coursera_sqlite%20(1).ipynb)

Build an Interactive Map with Folium

Markers - Markers are those points on the graph which indicate the coordinates or location of the launch site. These markers are created by executing the function `folium.marker()`.

Lines - Lines were drawn to connect the launch site with the locations and proximities by using the function `folium.polyline()`. These lines show the geographical relationship between the location and launch sites. These lines allow us to recognize geographical patterns.

Circles - Circles represent the proximity area around the launch site. These circles indicate the area that may influence the launch. These circles may help to create a safety zone or boundary. These proximity zones are created by the function `folium.Circle()`.

Build a Dashboard with Plotly Dash

Plots and Interaction in the dashboard make the visualization more easy to comprehend and carry out analysis -

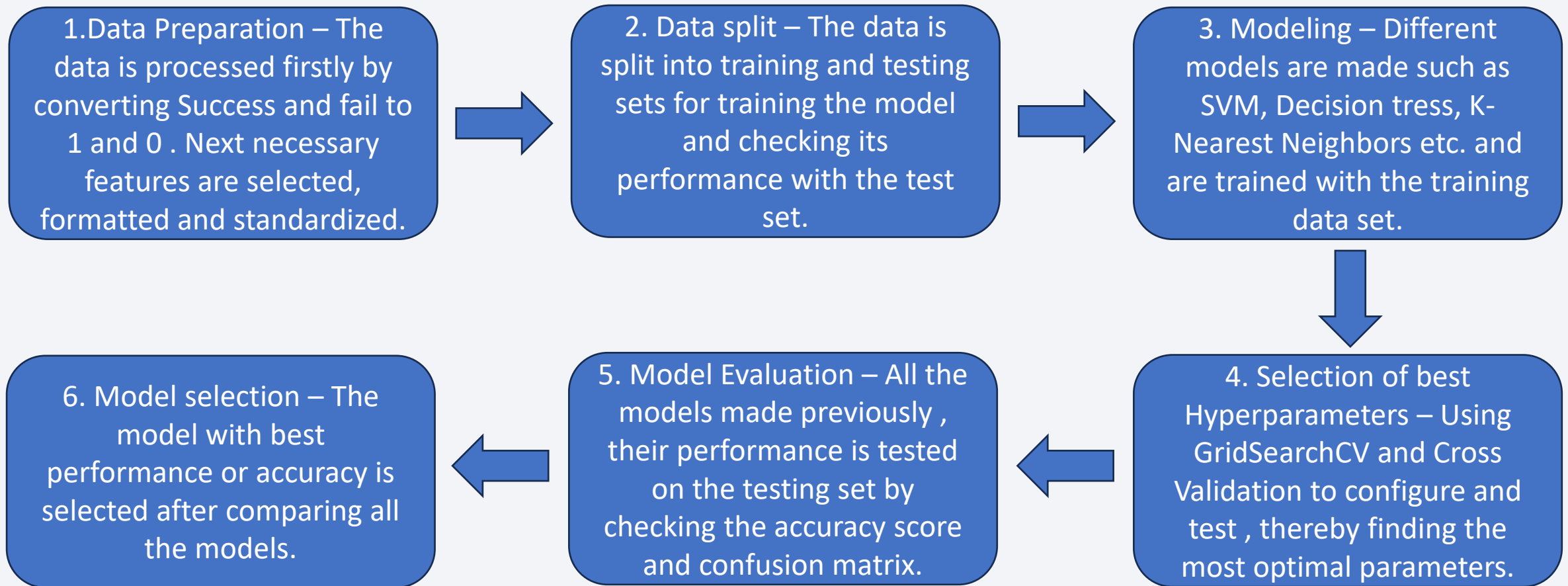
Pie chart – Pie chart was used to show the proportion of successful landings for all sites as well as the proportion of successful and failed launches for each site. It shows the overall launch success distribution.

Scatter Plot – The scatter plot was used to show the relation between the payload mass and launch success with the points indicating the launch outcome and type of booster.

Dropdown menu – The dropdown menu allows to select between all the site or each individual site. This dropdown facilitated individual site analysis as well as comparison between different sites

Slider for Payload – The slider allows the user to adjust payload mass. It allows the user to restrict the points or data belonging to a particular payload mass value. This helps in finding the optimal range of payload mass.

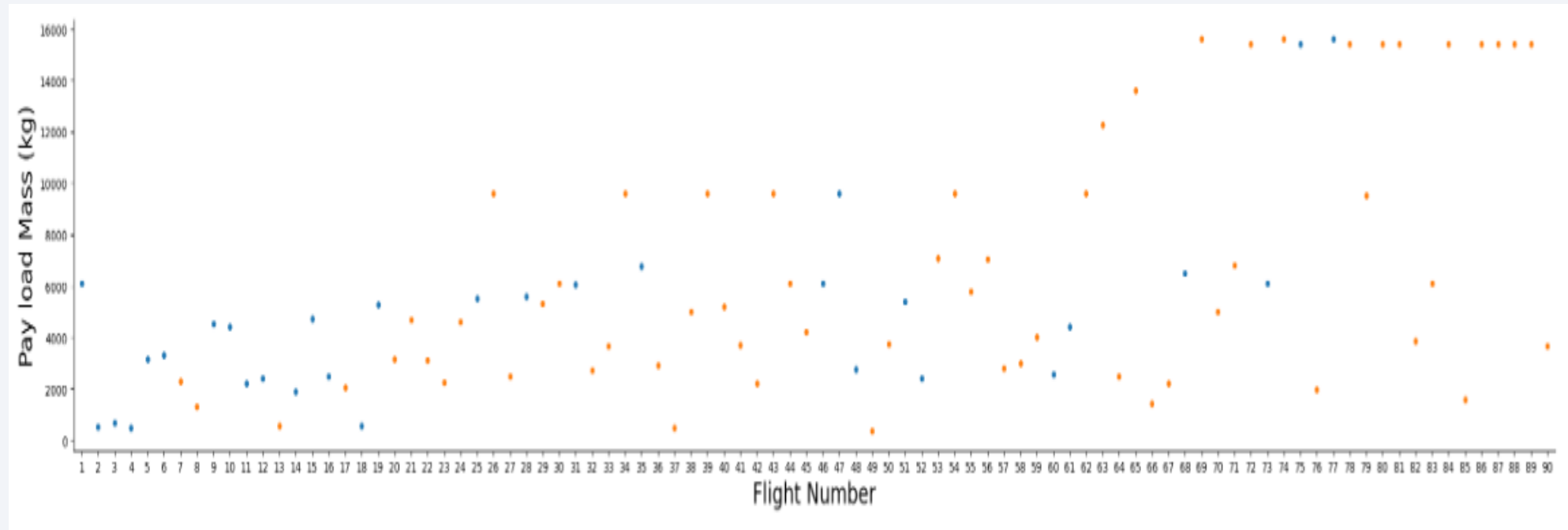
Predictive Analysis (Classification)



URL: [https://github.com/hamzaqdas7-design/IBM-final-project/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5%20\(1\).ipynb](https://github.com/hamzaqdas7-design/IBM-final-project/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5%20(1).ipynb)

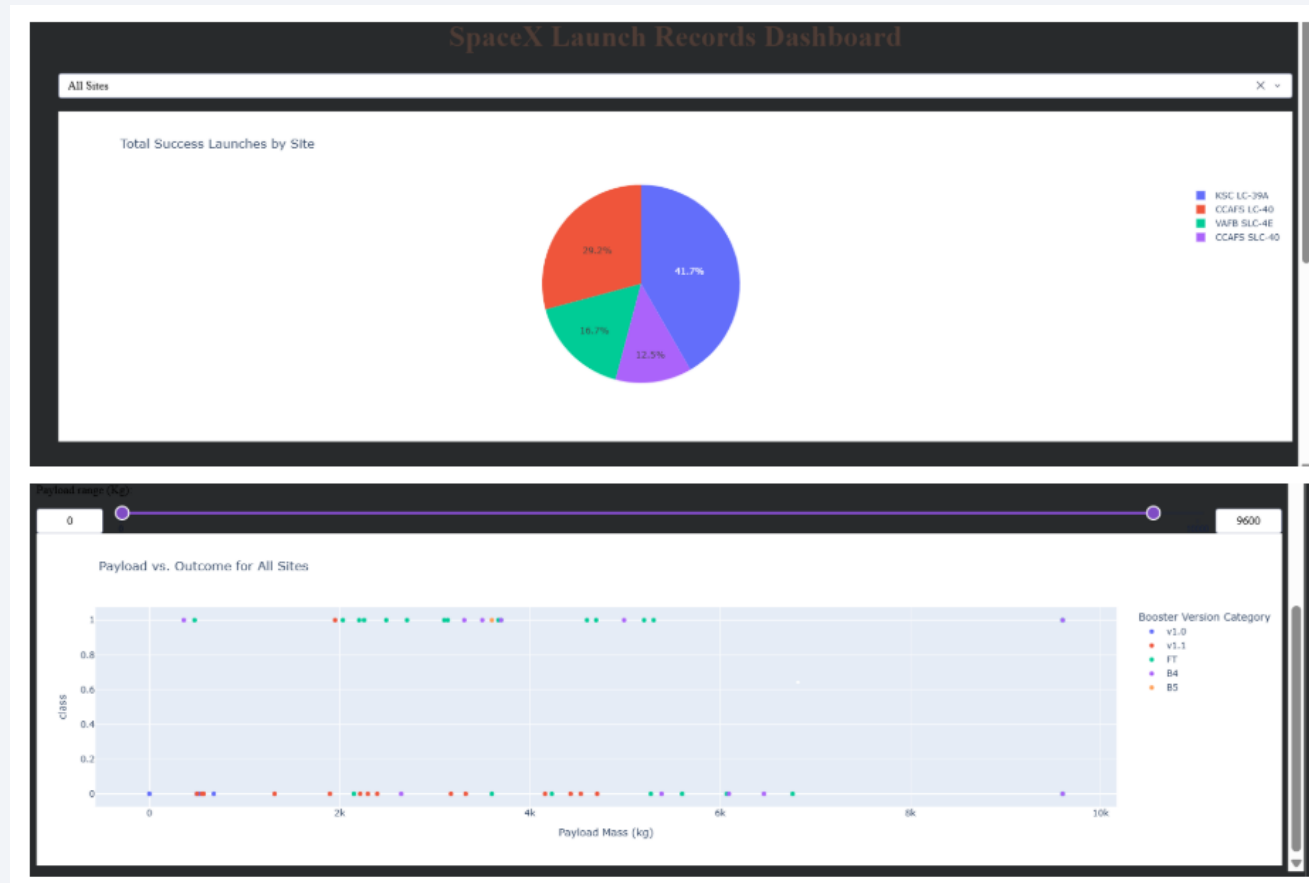
Results

Exploratory data analysis results -



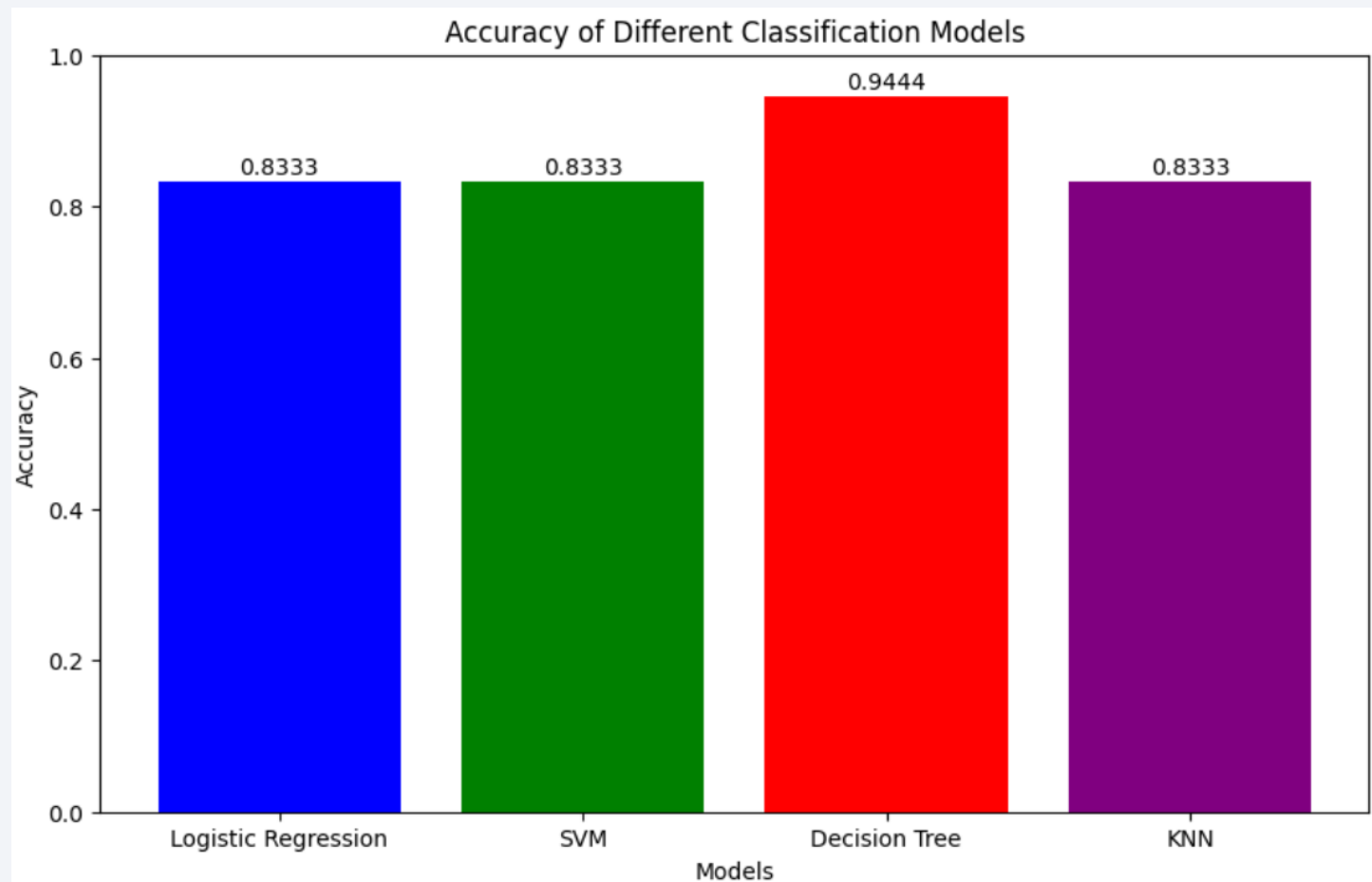
Results

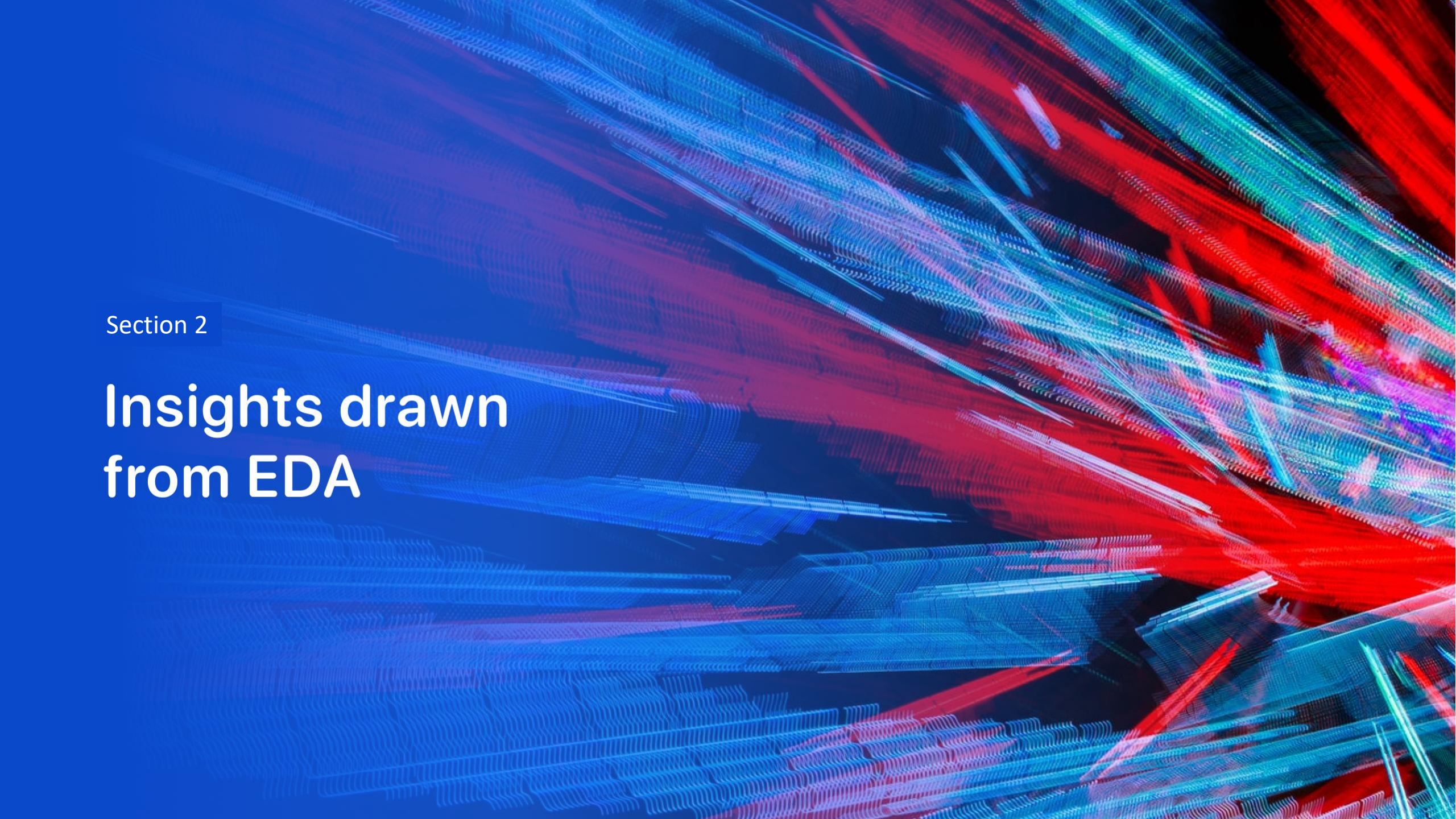
Interactive analytics demo in screenshots –



Results

Predictive analysis results -



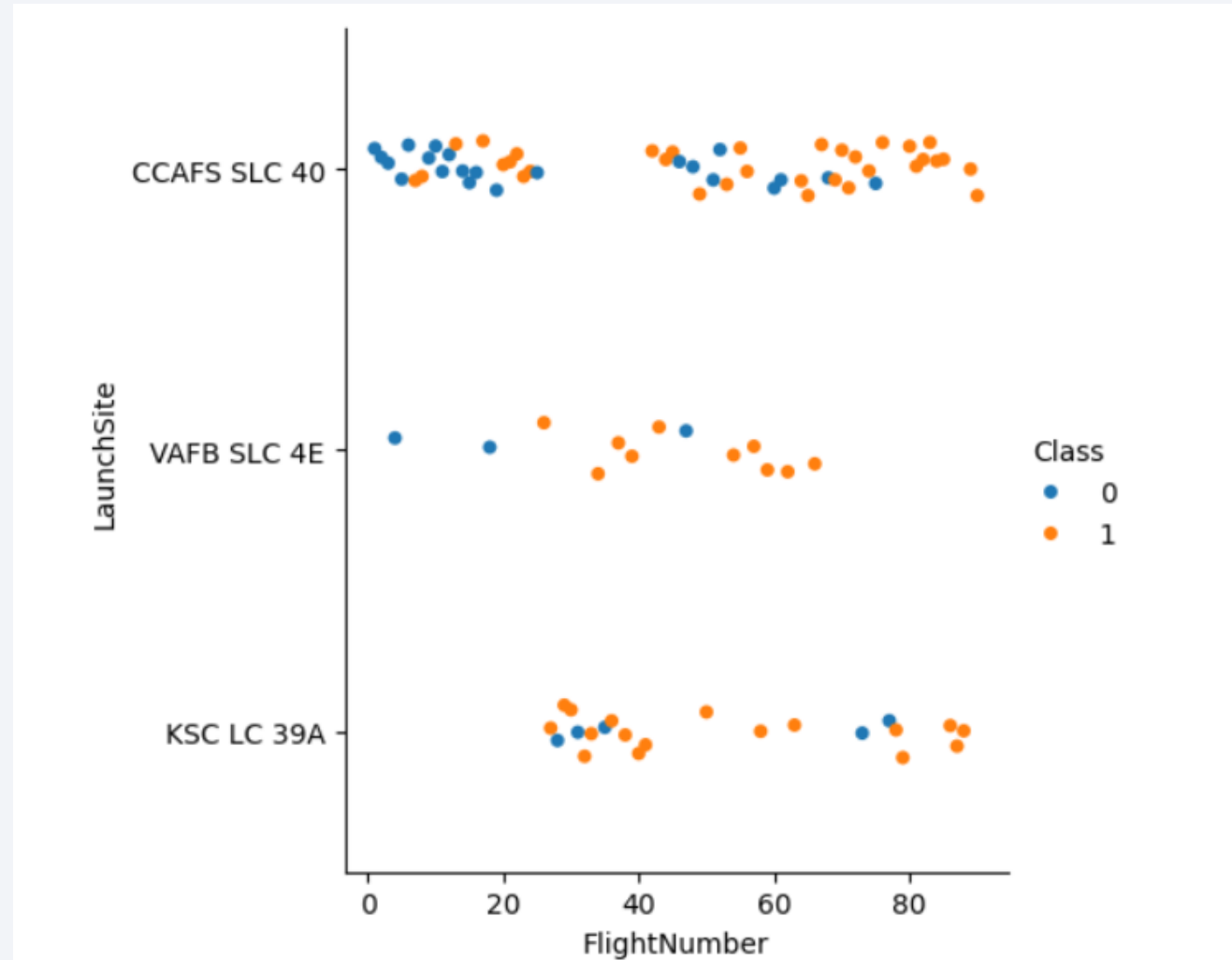
The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the upper right quadrant. The overall effect is dynamic and technological.

Section 2

Insights drawn from EDA

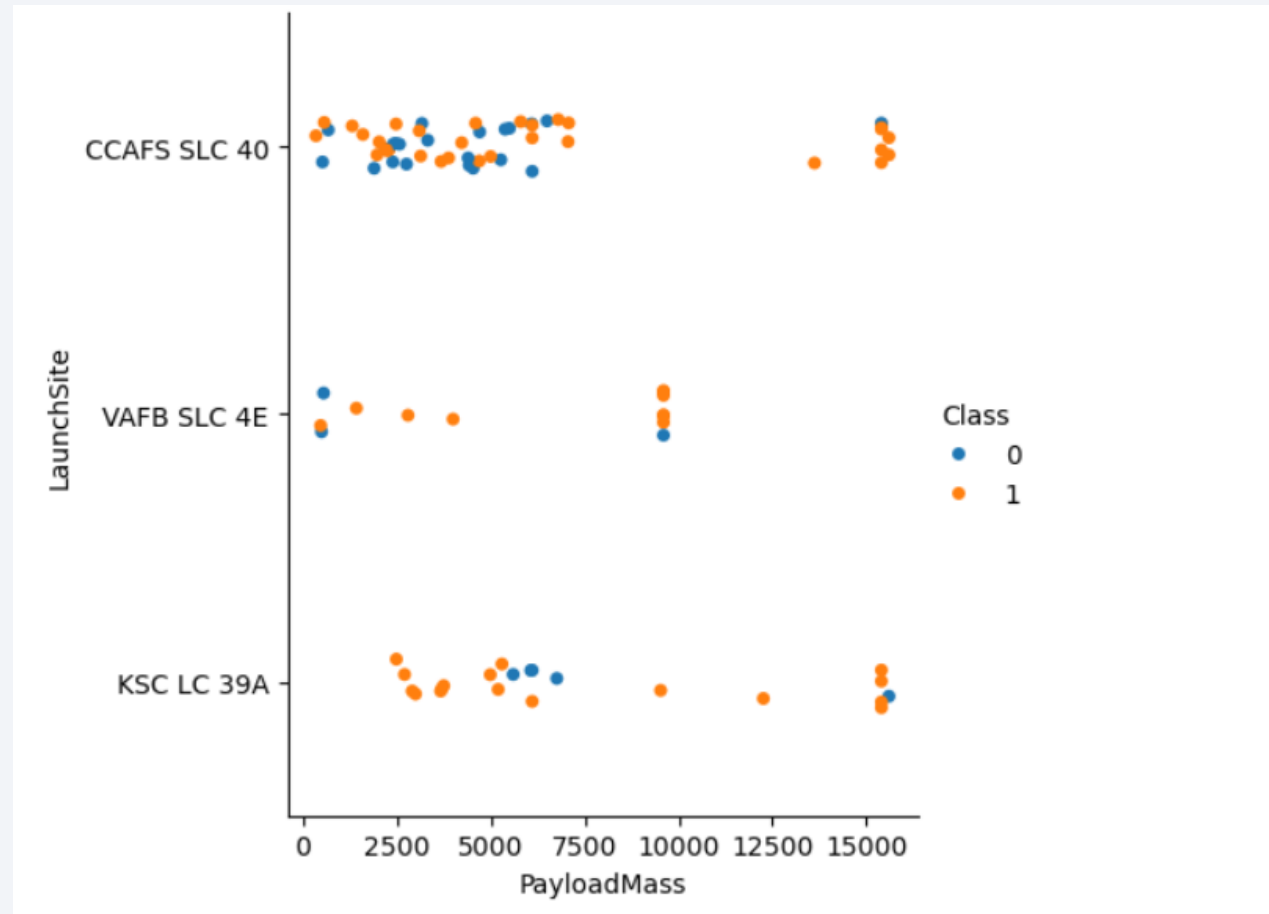
Flight Number vs. Launch Site

1. Mixed outcome can be seen in launch sites such as CCAFS SLC 40 and KSC LC 39A which indicate there are other reasons that influence the launch success.
2. Success and failures of launches are spread across all the launch. Thus, there is no clear trend of the success increasing or decreasing.
3. From this graph, it can be analyzed that some sites were used more intensively than others such as CCAFS SLC 40 was the most used in comparison to VAFB SLC 4E which was least used.



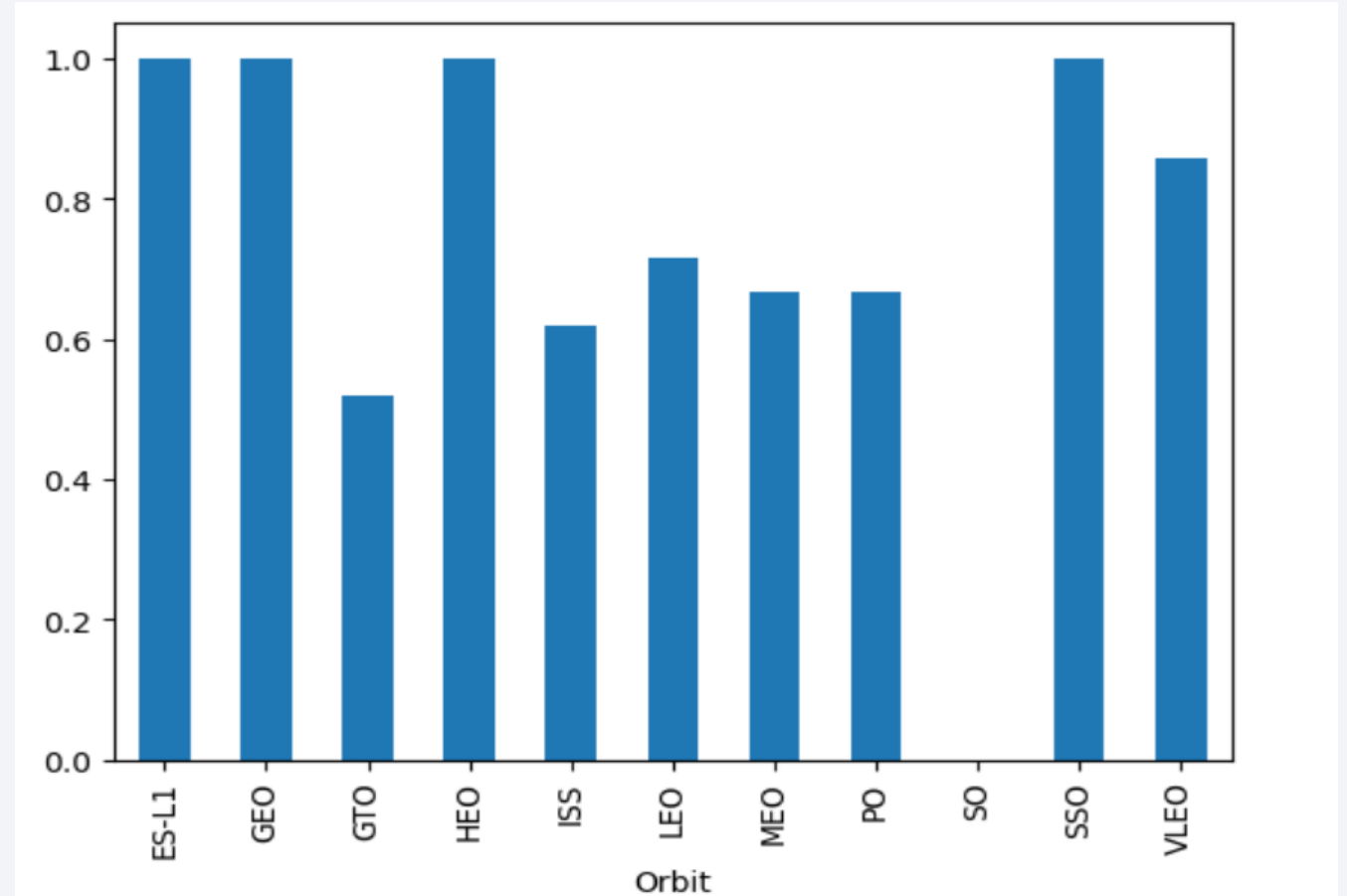
Payload vs. Launch Site

1. A wide range of payload masses is handled by different sites. It can be observed that CCAFS SLC 40 is the most used site and therefore has the biggest distribution. From CCAFS SLC 40 data, we can notice that it has the most sub-10000 kg launches as well as above 15000 kg launches.
2. The following graph also indicates that the launches observed from KSC LC 39A carry approximately 10000 kg of Payload mass



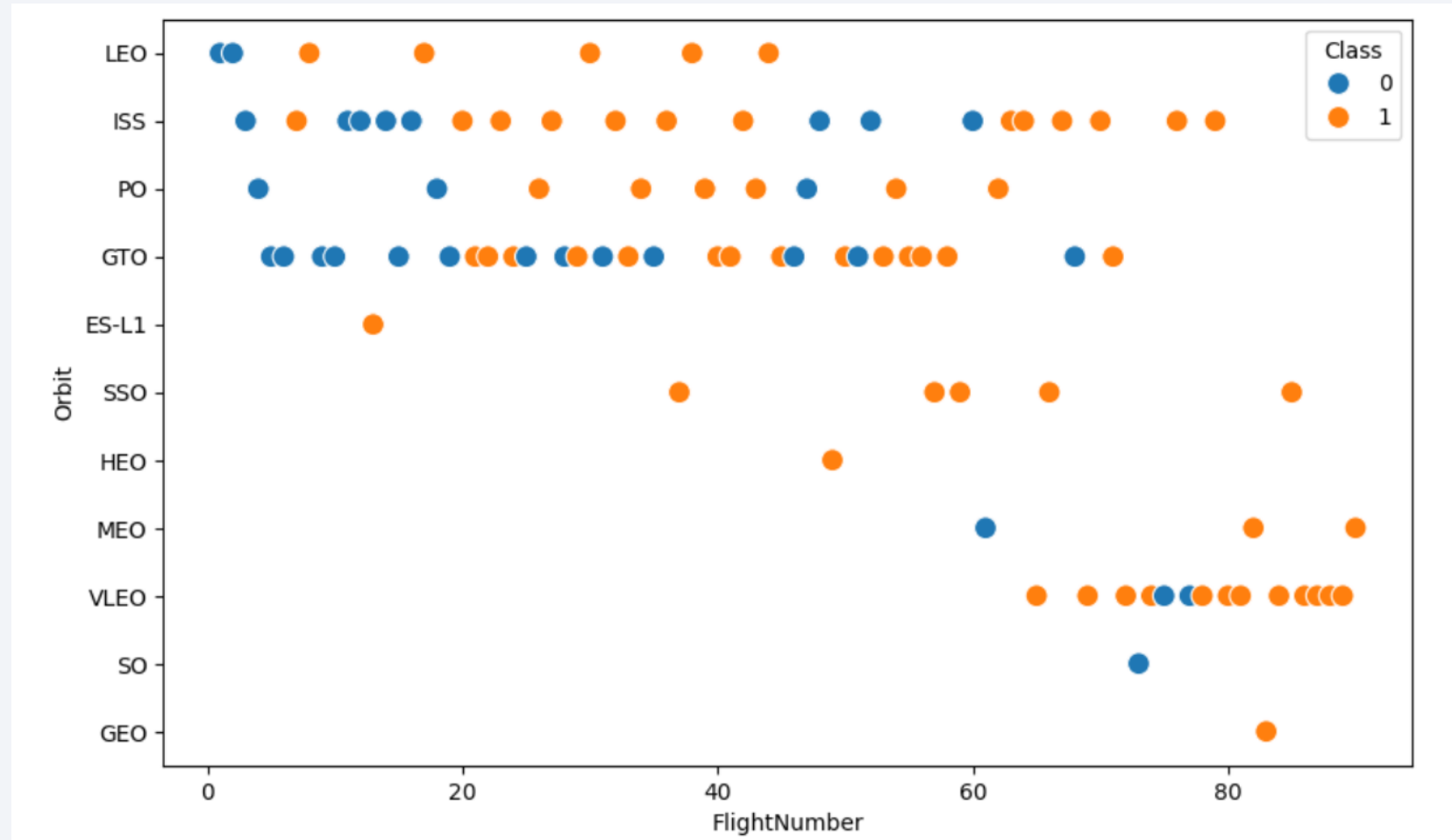
Success Rate vs. Orbit Type

1. Success rates of across all the orbit types vary. This shows that there is a relationship between the orbit type and success of the launch.
2. It can be analyzed that, the orbits ES-L1 , HEO, SSO, GEO have achieved a perfect success rate. This indicates that these orbits are reliable for launches.
3. Similarly, the GTO orbit has the lowest success rate. This may show that any launches to this orbit may face challenges.



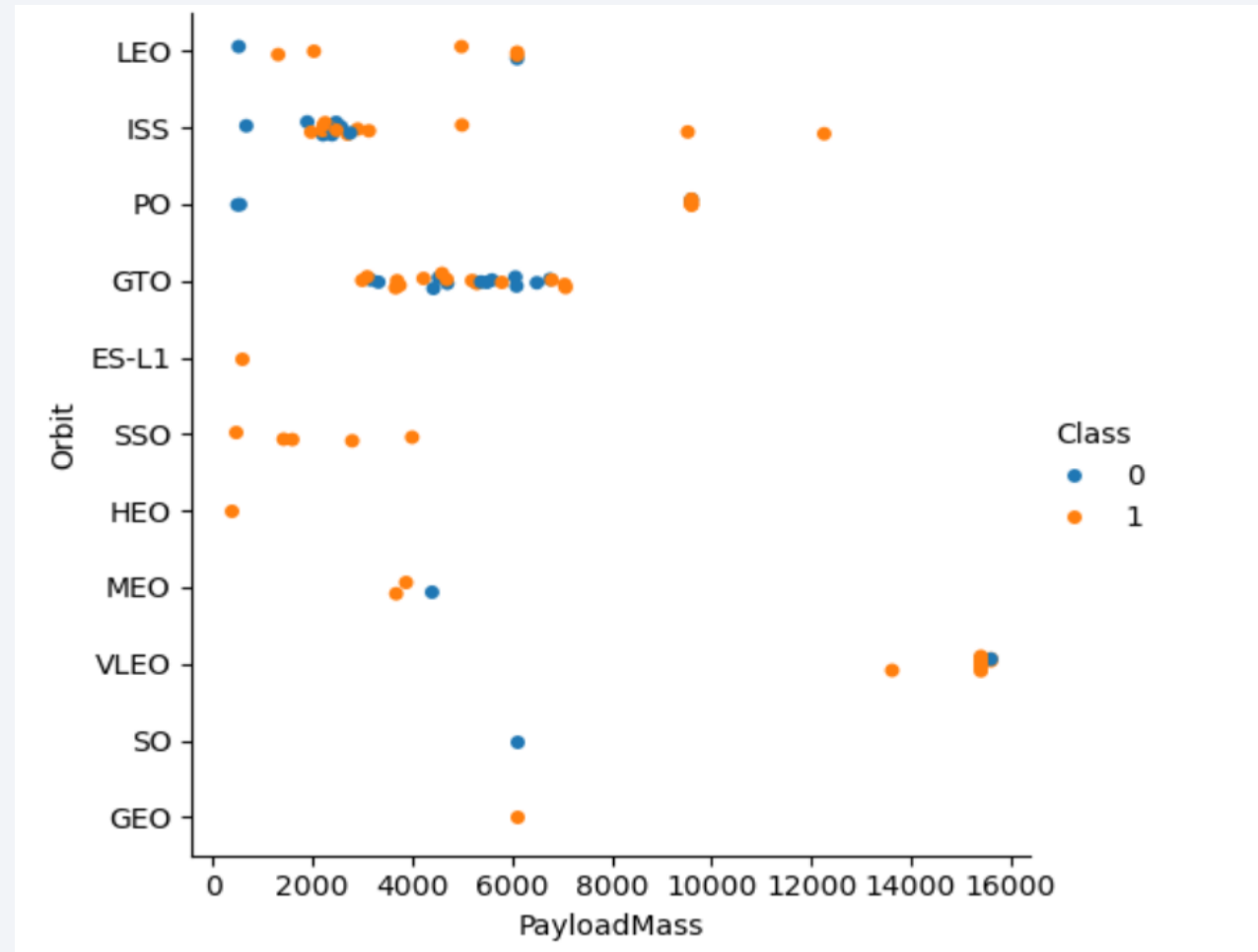
Flight Number vs. Orbit Type

1. From the following scatter plot graph, it can be analyzed that success of the launch site increases as time progresses. This shows that experience gained and improvements made to Falcon 9 help in the success of its launches. For example the ISS orbit showed mixed outcome in early launches, but later we can see more reliable launches.
2. This also shows that most of the failed launches occur in the early flights, reflecting SpaceX development.



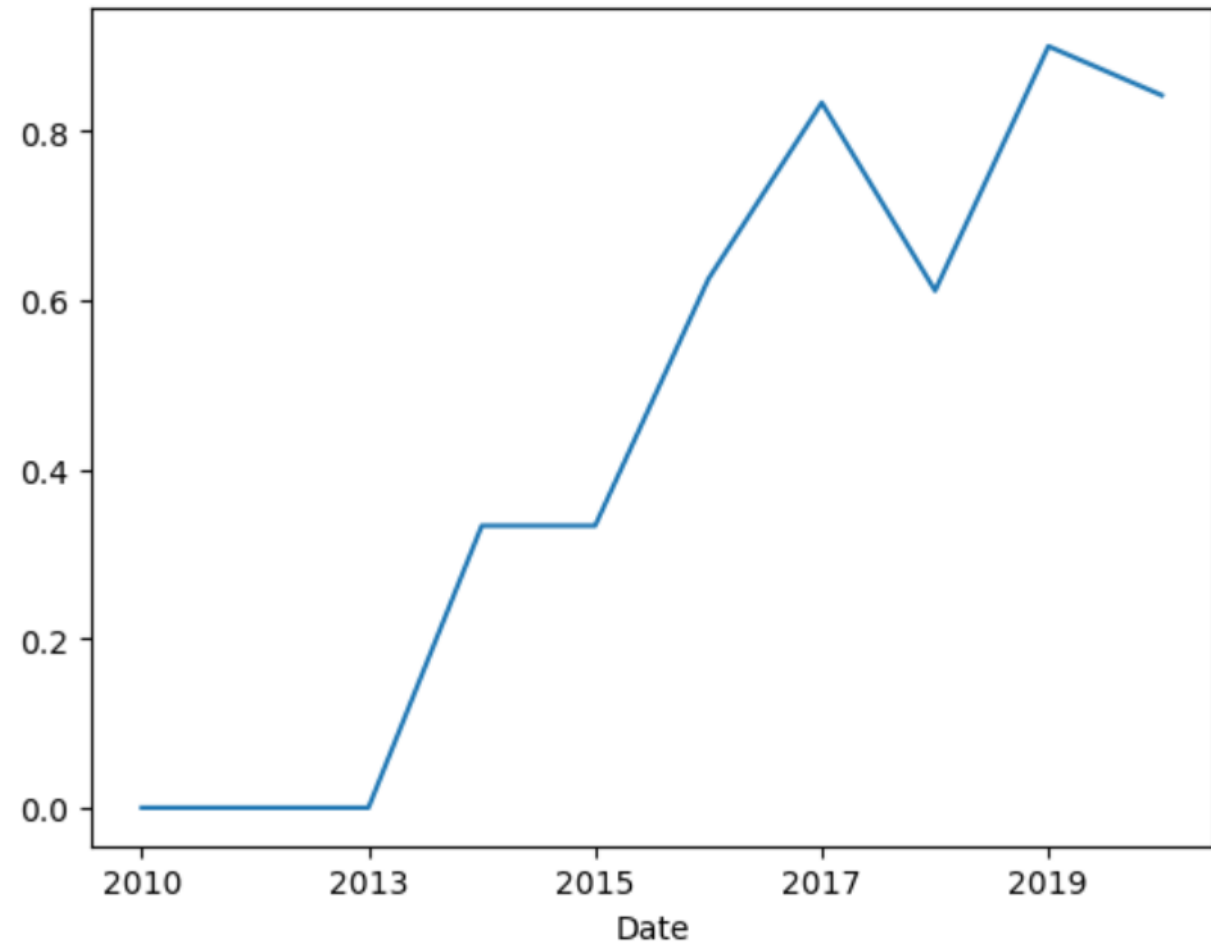
Payload vs. Orbit Type

1. The following scatter plot shows that payload mass varies across all the orbit types. Most of the successful launches lie around the payload mass of 6000 kg. Also, the GTO orbit shows a mixed outcome especially around this 6000 kg mark and thus it carries a moderate risk.
2. Heavy payloads ($\geq 13,000$ kg) correlate with launch success, while failures are concentrated in lighter payload missions (6,000 kg to 7000).



Launch Success Yearly Trend

1. The annual launch success rate has shown massive improvement from 2013 onwards, reaching approximately 80% by 2020. Despite a significant dip in 2018, the overall trend indicates increasing reliability, consistency and success in Falcon 9 launches over the years.
2. This overall increase in trend also showcase SpaceX ability to grow and develop and make improvements to their rocket.



All Launch Site Names

Retrieves all unique launch site from SPACEXTBL table -

```
In [30]: %%sql SELECT DISTINCT "Launch_Site"
          FROM SPACEXTBL;
```

```
* sqlite:///my_data1.db
Done.
```

```
Out[30]: Launch_Site
```

```
CCAFS LC-40
```

```
VAFB SLC-4E
```

```
KSC LC-39A
```

```
CCAFS SLC-40
```

Launch Site Names Begin with 'CCA'

Displays 5 records from the launch sites data with launch sites beginning with 'CCA' -

In [32]:

```
%%sql
SELECT *
FROM SPACEXTBL
WHERE "Launch_Site" LIKE 'CCA%'
LIMIT 5;
```

```
* sqlite:///my_data1.db
Done.
```

Out[32]:

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

Calculate the total payload mass carried by booster which were launched by NASA (CRS) -

```
In [34]: %%sql
SELECT SUM("Payload_Mass__KG_") AS TotalPayloadMass
FROM SPACEXTBL
WHERE "Customer" = 'NASA (CRS)';
```

```
* sqlite:///my_data1.db
Done.
```

```
Out[34]: TotalPayloadMass
         _____
                45596
```

Average Payload Mass by F9 v1.1

Average Payload Mass carried by booster of the Falcon 9 version 1.1 -

In [36]:

```
%%sql
SELECT AVG("Payload_Mass__KG_") AS AveragePayloadMass
FROM SPACEXTBL
WHERE "Booster_Version" = 'F9 v1.1';
```

```
* sqlite:///my_data1.db
Done.
```

Out[36]:

AveragePayloadMass

2928.4

First Successful Ground Landing Date

The date when the first successful landing was made on the ground pad -

```
In [37]: %%sql
SELECT MIN("Date") AS FirstSuccessfulLanding
FROM SPACEXTBL
WHERE "Landing_Outcome" = 'Success (ground pad)';
```

```
* sqlite:///my_data1.db
Done.
```

```
Out[37]: FirstSuccessfulLanding
          2015-12-22
```

Successful Drone Ship Landing with Payload between 4000 and 6000

Names of the boosters which have had success in drone ship landing while carrying a payload mass between 4000 to 6000 kg -

```
In [40]: %%sql
SELECT "Booster_Version"
FROM SPACEXTBL
WHERE "Landing_Outcome" = 'Success (drone ship)'
      AND "Payload_Mass_KG_" > 4000
      AND "Payload_Mass_KG_" < 6000;
```

```
* sqlite:///my_data1.db
Done.
```

```
Out[40]: Booster_Version
```

```
F9 FT B1022
```

```
F9 FT B1026
```

```
F9 FT B1021.2
```

```
F9 FT B1031.2
```

Total Number of Successful and Failure Mission Outcomes

Total number of success and failures -

```
In [41]: %%sql
SELECT "Mission_Outcome", COUNT(*) AS Total
FROM SPACEXTBL
GROUP BY "Mission_Outcome";
```

```
* sqlite:///my_data1.db
Done.
```

```
Out[41]:
```

Mission_Outcome	Total
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

List the names of the booster which carry the maximum payload mass -

```
In [43]: %%sql
SELECT "Booster_Version"
FROM SPACEXTBL
WHERE "Payload_Mass_KG_" = (
    SELECT MAX("Payload_Mass_KG_")
    FROM SPACEXTBL
);
```

```
* sqlite:///my_data1.db
Done.
```

```
Out[43]: Booster_Version
```

F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

Records displaying the month names, failure of landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015 -

```
In [26]: %%sql
SELECT
    substr("Date", 6, 2) AS Month,
    "Landing_Outcome",
    "BoosterVersion",
    "LaunchSite"
FROM SPACEXTBL
WHERE "Landing_Outcome" LIKE 'Failure (drone ship)%'
AND substr("Date", 0, 5) = '2015';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
Out[26]:
```

	Month	Landing_Outcome	BoosterVersion	LaunchSite
01	Failure (drone ship)	BoosterVersion	LaunchSite	
04	Failure (drone ship)	BoosterVersion	LaunchSite	

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Rank the count of landing outcomes such as success or failure between the date 2010-06-04 and 2017-03-20, in descending order -

```
In [27]: %%sql
SELECT "Landing_Outcome", COUNT(*) AS Count
FROM SPACEXTBL
WHERE "Date" BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY "Landing_Outcome"
ORDER BY Count DESC;
```

```
* sqlite:///my_data1.db
Done.
```

```
Out[27]:
```

Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

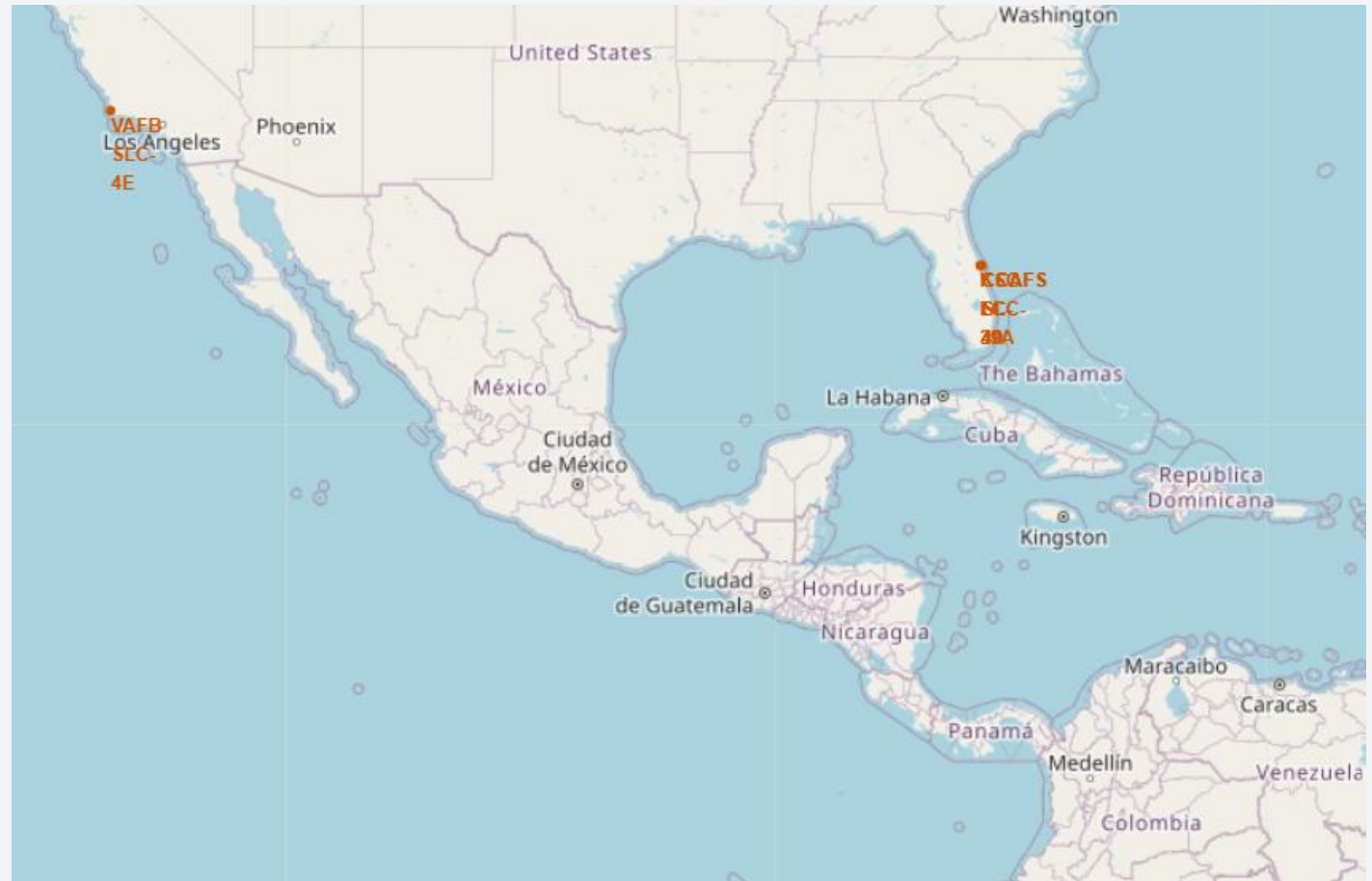
A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

Launch Sites Proximities Analysis

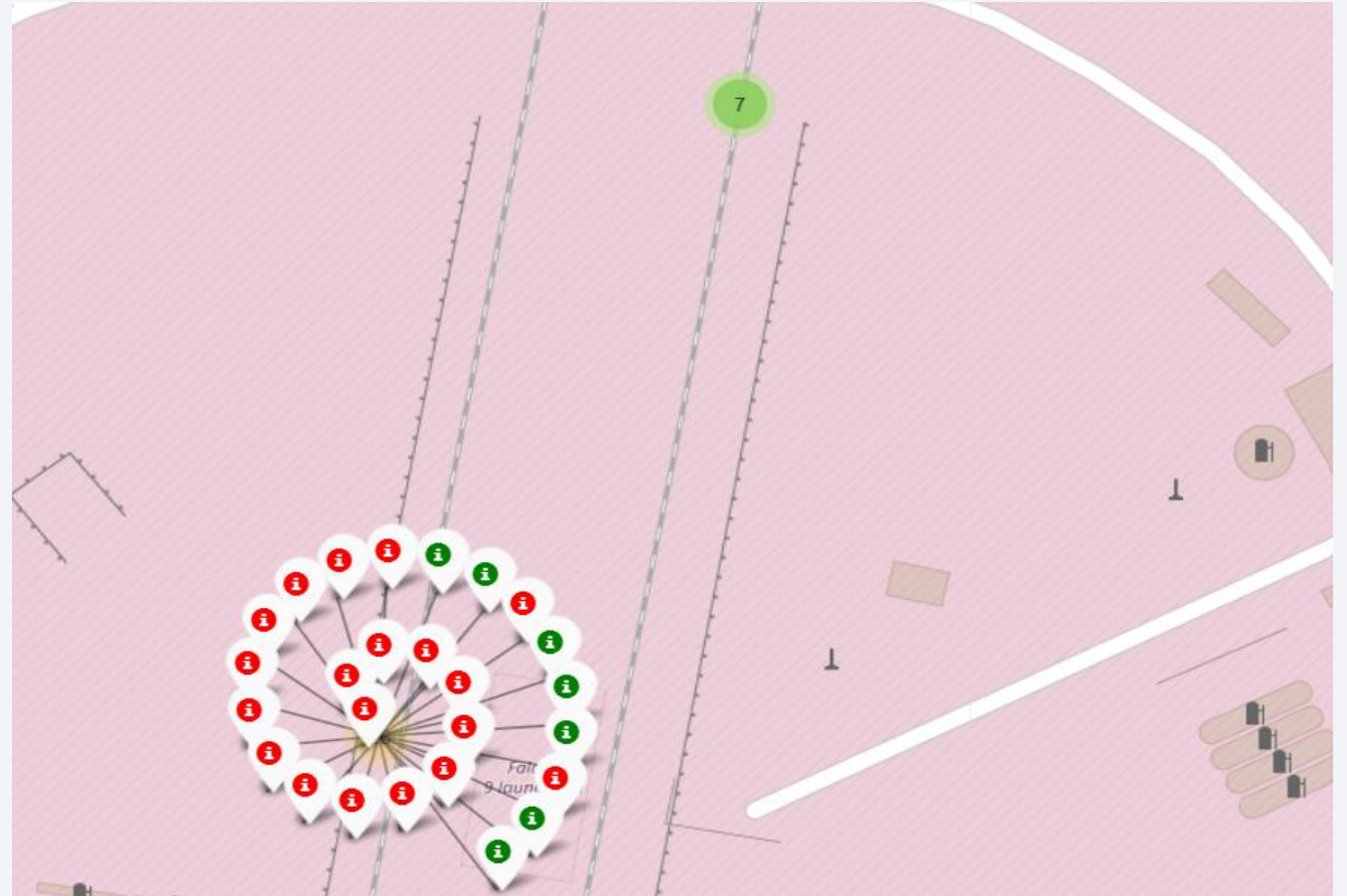
Markers of Launch Sites on Map

1. **Closer to Coastline** - All of the launch sites used by SpaceX are located near to the coast. This is especially advantageous for all the rocket trajectories as these launches can safely extend over the ocean. This helps in maintaining the safety of the nearby areas.
2. **Distance from Equator** - Most launch sites are closer to the equator, except VAFB SLC-4E which has a higher latitude of 34.63 and CCAFS SLC-40 is the most closes to the equator among the sites.
3. Also, it can be noticed that all the launch sites are located in USA, in its west and east coast respectively.



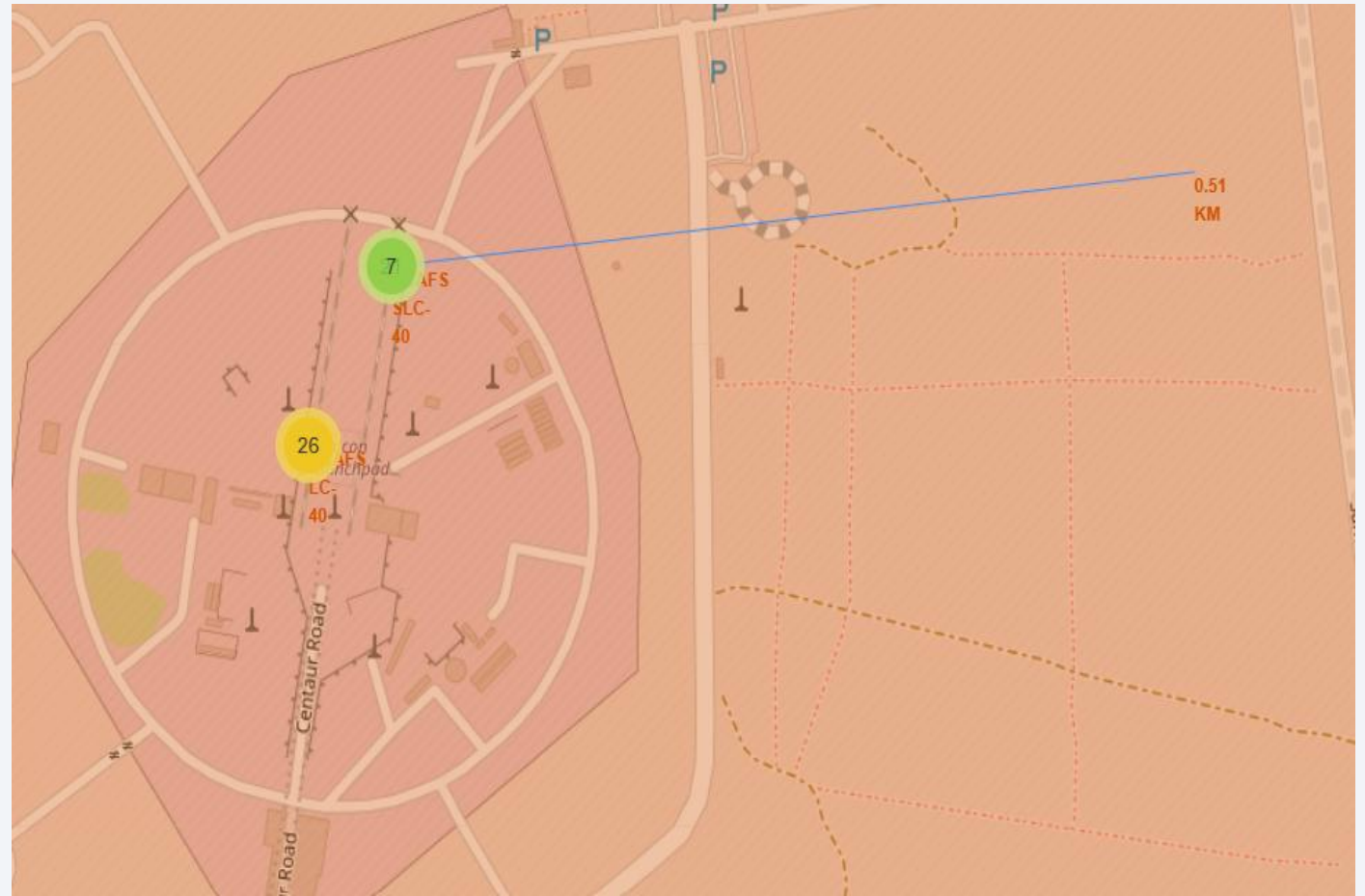
Successful and Failed Launches

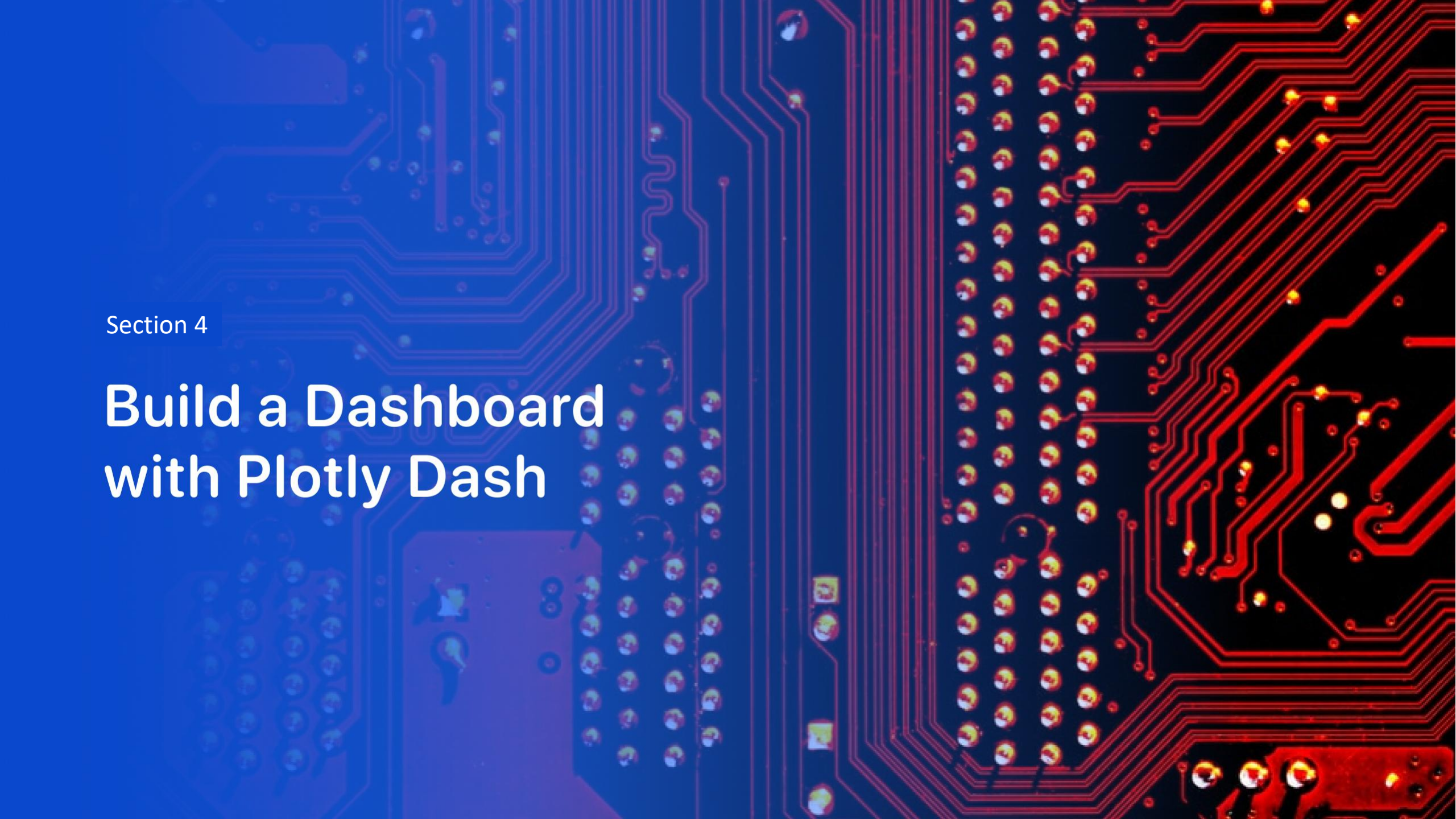
1. The following pictures shows the launch outcomes of the site CCAFS SLC-40. From the site we can clearly see that the red markers show that the launch was a failure and the green markers show that launch was a success. There are 19 red markers and 7 green marker. This visualization gives a clear picture about the performance of the launches.
2. The marker cluster is used to denote records or launches associated with each site
3. This spiral shape is known as spiderfly which is used by folium to show each distinct launches without overlapping



Distance of Launch Sites to its Proximates

The picture on the right shows the distance from the launch site CCAFS SLC-40 to the closest coastline point which 0.51 km far. This line to show the distance is made with the help of polyline. This proximity to the coastline is important as it help in over-sea trajectories of launches, operational recovery of the rocket in the sea as well maintaining public safety and the nearby areas.





Section 4

Build a Dashboard with Plotly Dash

Launch Success for All Sites

Total Success Launches by Site



The results of the pie chart are: KSC LC-39A - 41.7%, CCAFS LC-40 – 29.25%, VAFB SLC-4E – 16.7%, CCAFS SLC-40 – 12.5%. Thus, from the observations we can see that KSC LC-39A has the highest percentage of successful launches and this is the best and most reliable to site to use for launches.

KSC LC-39A Pie Chart

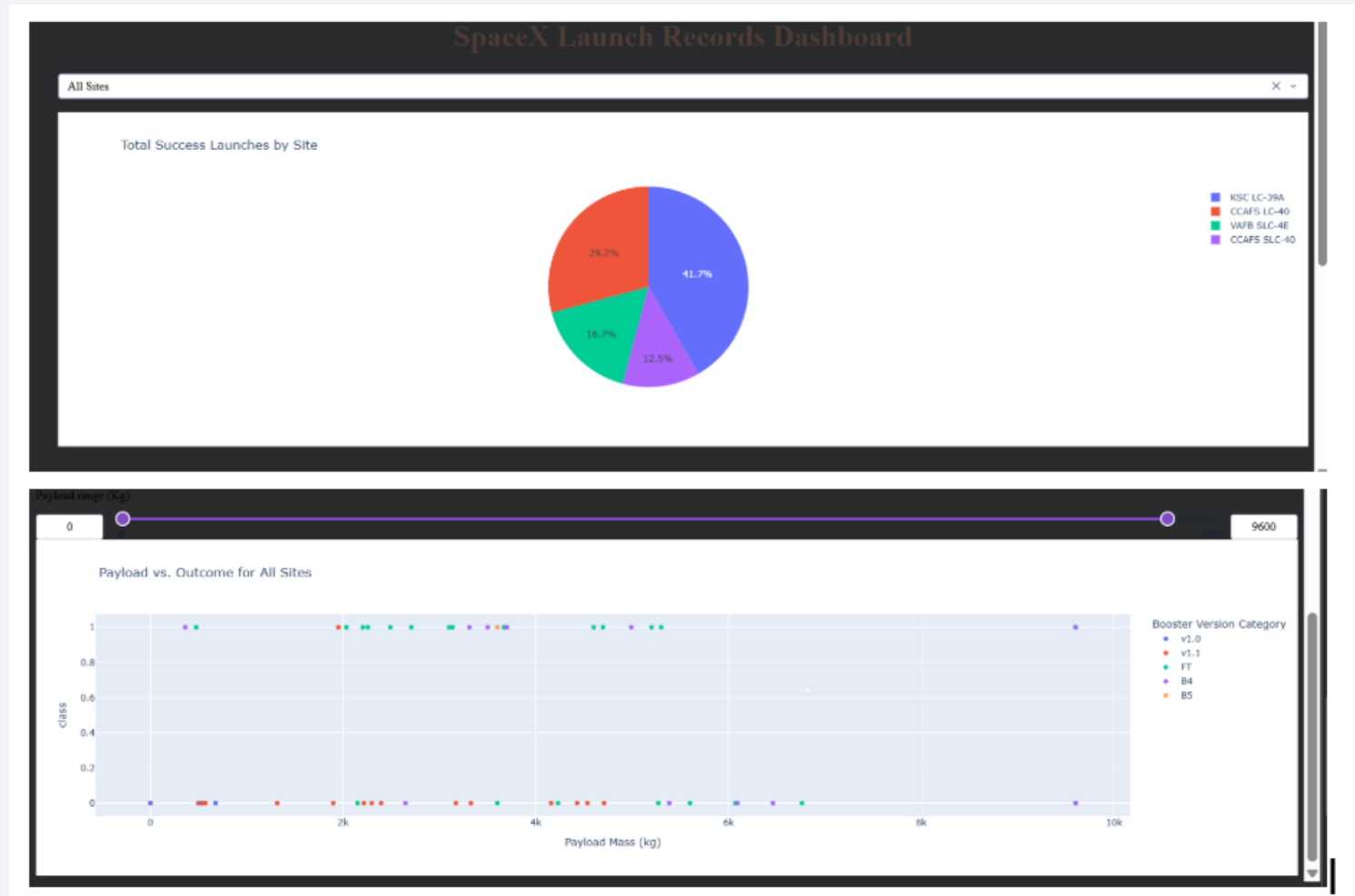
Total Success Launches for site KSC LC-39A



The KSC LC-39A has achieved success 76.9% of the total launches and has failed only 23.1% of the total launches. The results of the site KSC LC-39A, make it very highly reliable and effective for future launches by SpaceX.

Takeaways from SpaceX Launch Dashboard

1. Launch Site Success Rate – From the pie chart, we can infer that the KSC LC-39A is the best and most reliable site with success rate of 41.7%.
2. Payload Mass – The payload mass slider allows the user to select the desired payload mass. After analysis, we can see that the booster FT has the best result especially between the range of 2000 to 5000 kg. Also, the booster v1.1, shows the least amount of success in relation to payload mass, specifically below 4000kg.

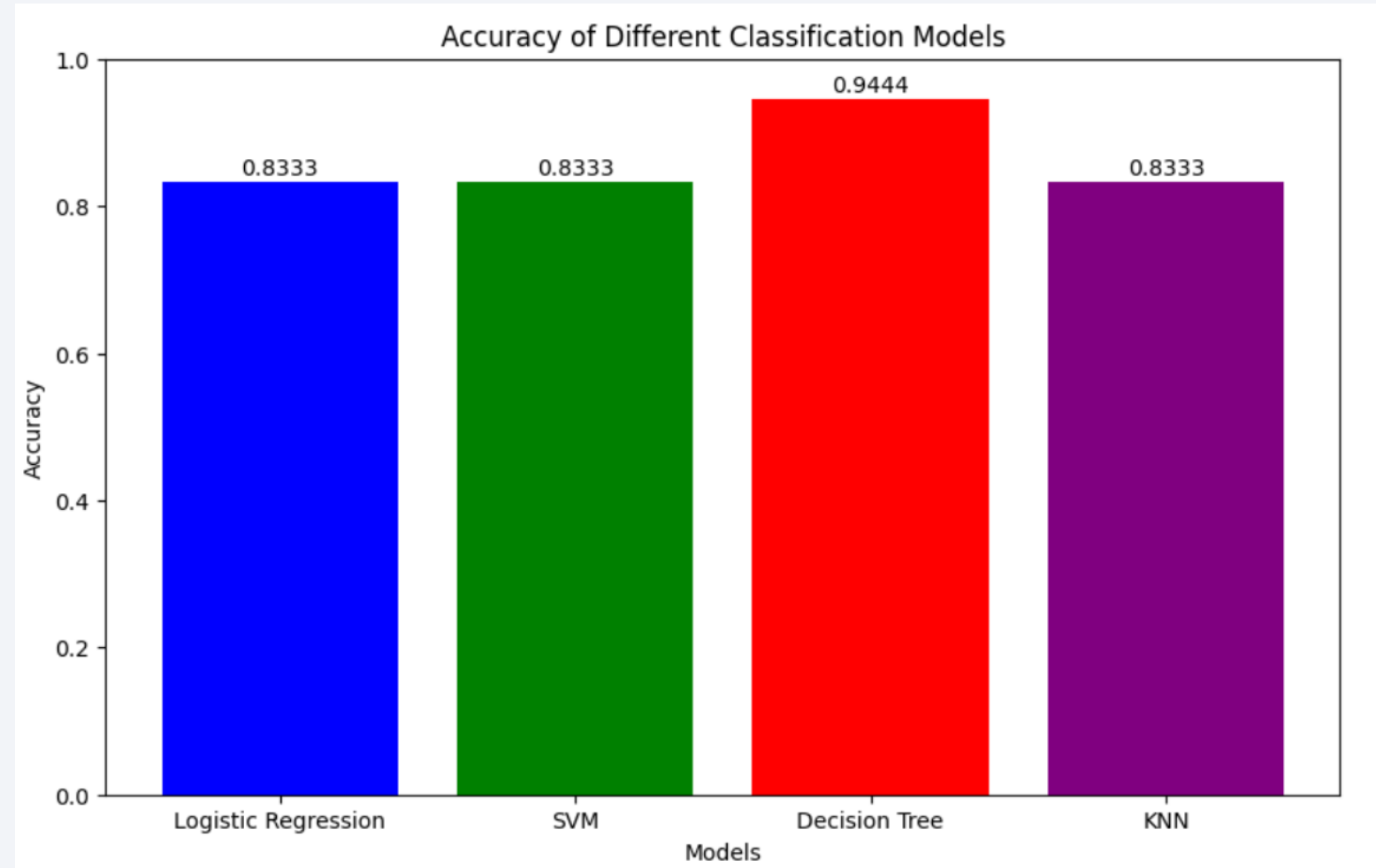


Section 5

Predictive Analysis (Classification)

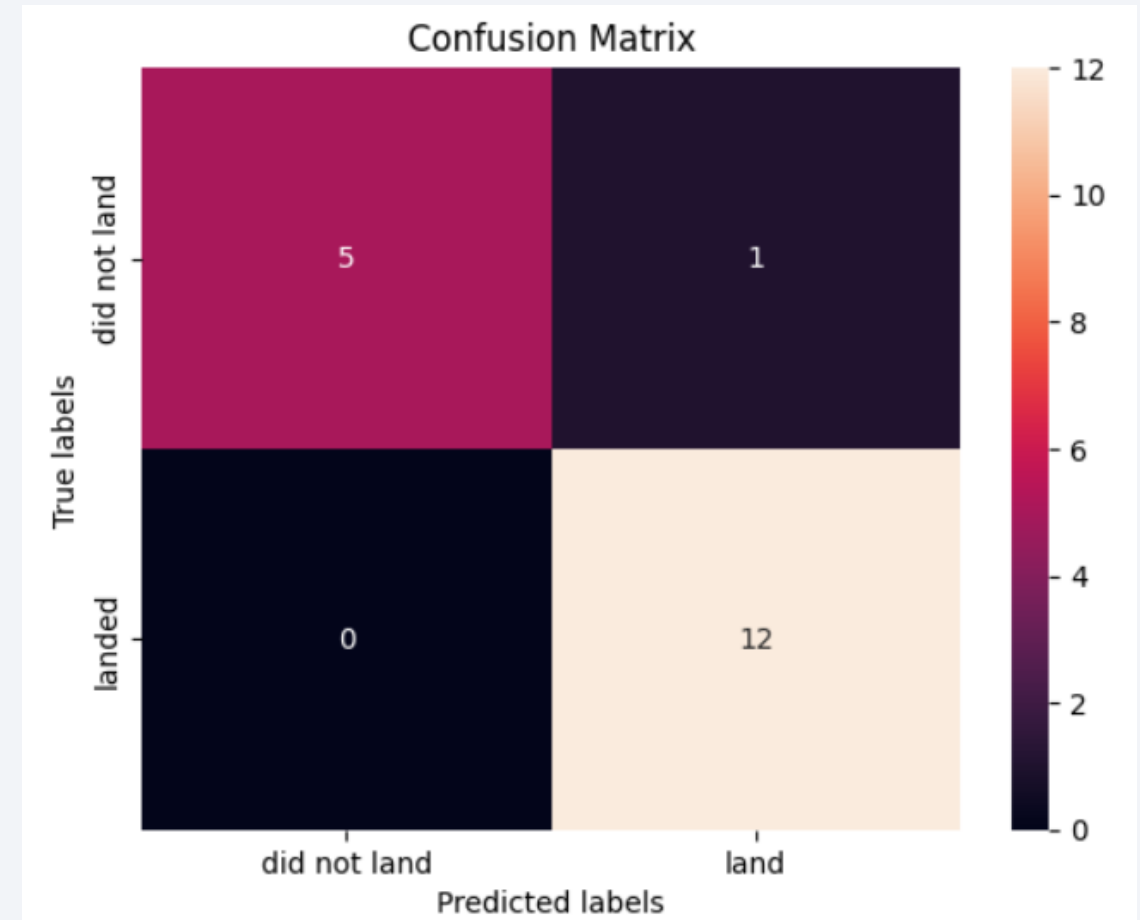
Classification Accuracy

From the following bar graph, we can infer that the Decision Tree model with an accuracy is of 94.44% is better than the other 3 models namely Logistic Regression, Support Vector Machines and K-Nearest Neighbors which all have an accuracy of 83.33. The Decision Tree model is the most effective algorithm and is the best at capturing the data's complex interactions.



Confusion Matrix

1. **Overall Performance** - The decision tree model has achieved an accuracy of 94.44%. Its precisions for the launches which landed, and which did not was respectively 92.3% and 83.33%.
2. **False Negative and False Positive** - The confusion matrix clearly shows that there are no false negatives, hence it has perfectly predicted each launch that has not landed. Coming to the false positive, there is one is 1 false positive. This indicates a slight approach towards over-optimism. Even though there is 1 false positive, but for the practical purpose this is a good score. In the aerospace industry this models shows reliability and good accuracy in predicting the landing of the rockets.
3. Lastly, this models shows a slight preference towards predicting launches landed and for the purpose of this industry it is of immense importance to determine the success of the landing



Conclusions

Point 1 - From the launch site performance, we can see that the site KSC LC-39A was the strongest performing launch site. This tell us that the location, trajectory, and nearby factors such as the ocean play an important role in the launch of the rocket and public safety.

Point 2 - The analysis of the different boosters showed us that the booster version FT displayed the best results for a wide range of payload mass. This demonstrate reliability and effectiveness of using the FT booster.

Point 3 - There was no clear pattern or trend where, higher payload mass lead to lower landing success. This suggest that there many other factors such as launch site and booster type which affect the success of the landing.

Point 4 - Prediction and Visualization – The machine learning model especially the Decision tree model which displayed an accuracy of 94.44% and predicted the landing outcome which is of paramount importance in the aerospace industry. Moreover, interactive visualization allows more detailed analysis and understand the spatial patterns. These visualization include dashboards and maps made with the dash and folium library respectively.

In conclusion, the predicted analysis conducted and interactive visualization made explore the factors that influence the success of the launch. Furthermore, the insights and results derived can help to enhance the launch strategies as well as develop the reusable rocket technology, thereby improving the aerospace industry.

Appendix

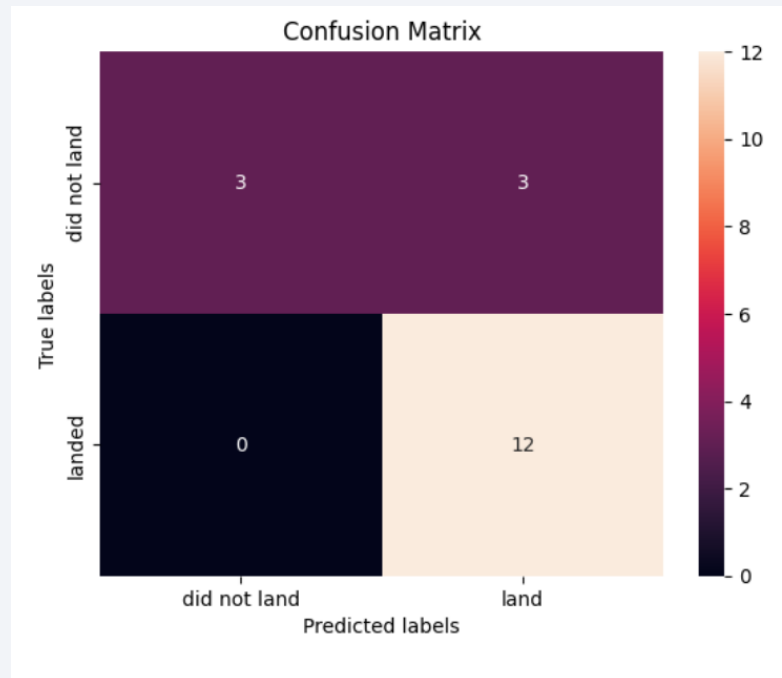
```
X_train, X_test, Y_train, Y_test
```

```
In [11]: X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.2, random_state=2)
```

we can see we only have 18 test samples.

```
In [12]: Y_test.shape
```

```
Out[12]: (18,)
```



Thank you!

